Jiawei (Marco) Guo

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Education

University of Washington, Paul G. Allen School Seattle, Washington

Bachelor of Science in Computer Science & Informatics | GPA: 3.92/4.0 Sept 2023 – May 2027

- Coursework: Cloud Computing, Machine Learning, Database Internals, Statistical Methods, Data Structure

Skills

Programming: Python (pandas, NumPy), SQL, Bash/Shell, R, MATLAB

Data Processing: ETL pipelines, data modeling, PySpark, Databricks, MySQL, MongoDB

Cloud & Distributed Systems: AWS, Azure, Oracle Cloud, distributed data workflows

Tools & Practices: Git, Linux, unit testing, Jupyter Notebooks

Experience

Data Engineer Intern, Molex – Lisle, IL June 2026 – Ongoing

- Analyzed blocked-order workflows using Celonis process mining , integrating enterprise data from SAP and Snowflake to identify process bottlenecks and uncover $2.3M in blocked orders, enabling operations teams to prioritize high-impact resolution opportunities.
- Built a supply chain project tracking platform on AWS, integrating Aurora PostgreSQL and Amazon Bedrock AgentCore to retrieve project progress, risks, blockers, and action items, giving executives a clear portfolio view and guiding leadership attention toward projects requiring immediate support.

Data Engineer Intern, Gekko Corporation – Seattle, WA Dec 2025 – May 2026

- Refactored existing AI platform into a multi-tenant PostgreSQL architecture, designing relational schemas and data isolation; improved query performance by 30%, reduced retrieval latency by 35%.
- Built automated analytics and monitoring workflows using OpenTelemetry and Splunk, processing highvolume interaction logs to detect anomalies and improve system reliability across 200+ users.
- Implemented validation and safety pipelines using NVIDIA NeMo Guardrails and model-based checks, reducing hallucination rate by 12% and non-compliant responses by 18% in production environments.

Research Assistant, Saveski Lab – Seattle, WA Sept 2025 – May 2026

- Built scalable data pipelines to process 2.3M+ Community Notes and 196M+ ratings across 20 shards, joining note, rating, and status data to enable large-scale analytics on hybrid human-AI fact-checking systems.
- Developed end-to-end NLP analytics workflows (BERTopic, M2M100, Transformers) to classify multilingual notes and align ratings with topic labels, uncovering distribution patterns and system-level behavior at scale.
- Conducted quantitative analysis across 3,340+ tweets, identifying that 97% of AI-generated notes remained unresolved and 49% underwent multiple revisions, revealing inefficiencies in collaborative moderation.

Teaching Assistant, University of Washington – Seattle, WA Sept 2024 – Ongoing

- Supported instruction for Machine Learning and Data Science Methods across 6 consecutive quarters.

Projects

New York Hospital Cost Prediction – Python, Tableau

- Built scalable data preprocessing and ETL pipeline to clean and transform 1.19M+ hospital records, enabling seamless Tableau integration in order to revealing care disparities between urban and rural hospitals.
- Developed and tuned Random Forest and two-stage XGBoost models, improving discharge prediction accuracy to 97% while reducing training time by 90% through target encoding.

GBI Global BI Pipeline – Databricks, PySpark, Snowflake, MongoDB, Power BI

- Built ELT pipelines on Databricks using PySpark to process and transform 19.6M+ Amazon review records from MongoDB, enabling scalable analytics over large datasets.
- Designed a cloud data warehouse schema in Snowflake and integrated it with distributed processing workflows.